B

3. This is a systematic random sample, considered to be reasonably unbiased.

4. Options (ii) and (iii)

5. (a) IV (b) III (c) II (d) V (e) I

6. (a) Neither is there matching of individuals across the two groups, nor is there before – after comparisons on the performance of individuals. This is not a matched pairs study.

   (b) Were employees self-selecting the procedure (old or new) they used? If so, the study is observational. If they were assigned procedures to use, then it is an experiment.

7. (a) iv (b) i (c) iii (d) ii

8. (a) bimodal, symmetric

   (b) median

   (c) 5, 9.5, 13.75, 18.5, 23
   $23 - 5 = 18$
   $18.5 - 9.5 = 9$
   } all of these are estimates

   (d) $q_3 + (1.5)(\text{iqr}) = 18.5 + (1.5)(9) = 32$

9. $\bar{x} = \frac{1}{4}(25 + 14 + 19 + 22) = \frac{1}{4}(80) = 20$

$$\text{Variance} = \frac{1}{3}\left[(25-20)^2 + (14-20)^2 + (19-20)^2 + (22-20)^2\right]$$
$$= \frac{1}{3}(25 + 36 + 1 + 4) = \frac{1}{3}(66) = 22$$
$$\Rightarrow \quad s = \sqrt{\text{variance}} = \sqrt{22} \doteq 4.69$$

10. (a) $P(X = 3) = 1 - (0.16 + 0.18 + 0.32 + 0.13) = 0.21$

    (b) $P(X \le 1) = P(X = 0) + P(X = 1) = 0.16 + 0.18 = 0.34$

    (c) $P(X = 1 \mid X \le 2) = \dfrac{P(X = 1 \text{ and } X \le 2)}{P(X \le 2)} = \dfrac{P(X = 1)}{0.16 + 0.18 + 0.32} = \dfrac{0.18}{0.66} = 0.2727$

    (d) $\mu_x = (0)(0.16) + (1)(0.18) + (2)(0.32) + (3)(0.21) + (4)(0.13)$
    $= 0 + 0.18 + 0.64 + 0.63 + 0.52 = 1.97$

11. $P(\text{Closed}) = P(\text{Snows and Closed})$

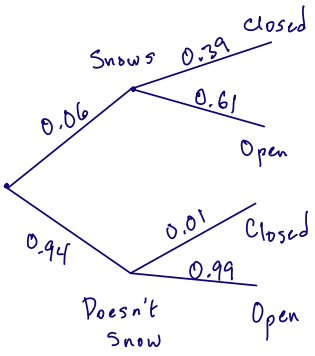
$+ P(\text{Doesn't Snow and Closed})$

$= P(\text{Snows})\,P(\text{closed} \mid \text{Snows})$

$+ P(\text{Doesn't Snow})\,P(\text{Closed} \mid \text{Doesn't Snow})$

$= (0.06)(0.39) + (0.94)(0.01)$

$= 0.0234 + 0.0094 = 0.0328$



12. Options (a), (c), (d) and (f)

13. (a) Willa's count X is binomial, as each time she draws another token, the probability of success (green) has been restored to its original value.

(b) $n = 8, \quad p = \frac{9}{22}$